

TFG Initial Report: Are AI models fooled by 3D illusions?



By: Ivan Martín Campoy

1. Problem Definition and Motivation

1.1. Motivation

The motivation for this project is that it covers two of my favorite fields: optical illusions and artificial intelligence. I have always been fascinated how the human brain can be tricked by 2D representations of 3D spaces; furthermore, given the recent progression of generative models lately, I saw an opportunity to test if machines are also susceptible to the same optical illusions.

In this project, I intend to conduct an analysis using AI foundation models to investigate whether they can “perceive” the logical impossibilities or if it is just following a pattern blindly.

Due to the highly visual nature of the project, the result allows to visualize them in a clear and understandable way, making the final conclusions accessible.

1.2. Problem statement and Research Question

The core of this project lies in the limitations of current AI vision foundation models to interpret spatial logic beyond the 2D pattern recognition.

While AI is used for spatial tasks, it remains unclear if these model can “understand” 3D spaces or they are just following pixel patterns. The main question for this TFG is: “How do Vision Foundation Models understand the different types of 3D optical illusions?”

2. Objectives and Contributions

The main objective is to evaluate the spatial intelligence, and structure consistency of AI vision foundation models using a diverse set of 3D optical illusions.

2.1. Specific Objectives

To achieve this, the following objectives have been defined:

1. Taxonomy dataset curation: Organize a specialized dataset of 3D optical illusions to test different failures in AI models.
2. Implementation of a Multi-generational Pipeline: Process the illusory data through the state-of-the-art AI models like the Depth Anything family to generate spatial outputs, such as depth maps or 3D meshes.
3. Spatial Integrity Assessment: Establish a method to extract and analyze depth values and geometric coordinates to verify the results.

4. Comparative analysis: Evaluate if the AI models replicate human cognitive biases (as defined by the Helmholtz-Gregory framework), or digital hallucinations when faced with geometric or other type of illusions or if they follow an algorithmic pattern of error.

2.2. Expected Contributions

1. Identification of Model Biases: Mapping natural image priors (such as light coming from above) dominant in modern ViT architectures.
2. Validation of Biological Inheritance in AI: Analyze whether modern models replicate human-like errors.

3. Context and State of the Art

3.1. Defining Optical illusions in the context of vision

To evaluate how artificial intelligence processes visual paradoxes, it is important to establish the definition of an optical illusion. According to Hermann von Helmholtz^[1], a visual perception is the unconscious inference the brain performs from the processed data using previous knowledge. Richard Gregory (1997) expanded this topic by explaining that these perceptions can be understood as a “predictive hypotheses”, this means that humans acknowledge the world by projecting these hypotheses as an immediate reality.

Having established this definition, we can conclude that an optical illusion is not a visual malfunction (for example, water reflecting light and deforming a signal), but a faulty processing due to the cognitive rules (priors) learned through previous experience that disagree with the physical world and their geometric laws. In the context of this TFG, illusions are used as a test for AI to reveal if the model’s depth estimation is based on geometric logic or only on the learned statistical priors.

3.2. Natural Image Priors

To prioritize human survival, the brain evolved into rapid processing over mathematical precision. These regularities, often called “priors” are based on assumptions such as the light usually coming from above or parallel lines converging at the horizon.

Hubel & Wiesel (1962) discovered that the brain’s primary visual cortex (V1)^[2] is optimized to detect specific orientations and edges occurring in the environment using ‘simple’ and ‘complex’ cells where simple cells act as basic edge detectors..

3.3. Taxonomy of Visual Paradoxes

To conduct a structured analysis, the dataset is organized into five different categories inspired by R. Gregory’s categories.

3.3.1. Impossible objects: Objects or structures that are locally coherent but globally impossible (e.g., Penrose triangle [\[Fig. 1, left\]](#), or Escher's cube [\[Fig. 1, right\]](#).

3.3.2. Forced perspective: Illusions designed to be viewed from specific fixed monocular viewpoint, leading to a scale, distance or alignment illusion. One common example is Ames room [\[Fig. 2\]](#).

3.3.3. Depth ambiguity: 2D Bistable projections that allow for multiple valid 3D interpretations, these illusions test the AI statistical bias checking which solution prefers to "see". E.g, Necker's Cube [\[Fig. 3\]](#).

3.3.4. Converging lines (linear perspective): It is based on classical line illusions like the ponzo illusion, the objective is to check the importance of the background context when computing object depth. Ex: corridor's illusion [\[Fig. 4\]](#).

3.3.5. Shading & illumination (Shape-from-shading): Evaluates how AI models interpret light and shadows to infer volume. It tests if the model interprets local luminance gradients as actual volume or simply as changes in 2D surface color. Ex: Gardner's Dragon [\[Fig 5.\]](#)

3.3.6. Note on Taxonomic overlap

It is important to acknowledge that the boundaries between these categories are often diffuse. In many instances, an illusion can rely on multiple deceptive mechanisms. One example could be an image of a person "holding the sun", as it can be classified as forced perspective due to the specific viewpoint alignment, but it also has a significant depth ambiguity since it defies the distance between objects and their sizes.

3.4. State of the Art

3.4.1. Foundational Perspectives: Psychology and Mathematics

Before evaluating these artificial systems, it is essential to establish some biological and geometric common rules used in optical illusory images.

Akiyoshi Kitaoka: The Psychology of Contrast and Latency

Akiyoshi Kitaoka is a pioneer in the study of peripheral drift and anomalous motion [\[3\]](#). His work focuses on how the human visual system prioritizes certain signals, such as high-contrast edges and luminance gradients over others. Specifically, research by Kitaoka, Conway et al. [\[4\]](#) shows that the primary visual cortex (V1) processes white and black (high-contrast information) faster than low-contrast (grays), causing a temporal lag. They follow a four-step luminance rule, the brain interprets the timing mismatch as motion, and therefore projects it as dynamic. By following the proposed sequence, the illusion's reaction is maximized.

The Luminance-Depth Connection

As explained before, one of the dataset's division include a shading section, where the luminance gradients are an important pillar to understand the results obtained from the foundation models.

Kokichi Sugihara: Geometric Impossibility and Mental Models

On the other hand, from a mathematical perspective, Kokichi Sugihara focuses on geometric frameworks to understand 3D volumes. One of his most famous examples is the “Ambiguous cylinders”, a 3D illusion that tricks the brain to perceive an object in an inconsistent way while viewing it directly and its reflection in a mirror. [\[5\]](#)

3.4.2. From CNNs to Foundation Models

The technological evolution of computer vision has evolved from task-specific architectures to a more general-purpose Vision Foundation Models. Convolutional Neural Networks (CNNs) were the standard, being used to extract features, however they sometimes missed the whole understanding of the scene since they prioritize local pixel regions.

The arrival of Vision Transformers (ViTs) caused a paradigm shift, thanks to their self-attention mechanisms they can weigh the importance of different parts of an image simultaneously, regardless of their distance, this allows a “global” understanding.

3.4.3. Current research in AI Perception: Diffusion and CLIP

With the arrival of Diffusion Models (DMs) and their unique denoising-based architecture, new ways of understanding illusions have been created. Recent research by Gomez-Villa et al. (2025)[\[6\]](#), demonstrates that diffusion neural networks can perceive brightness and color visual illusions like humans do, following statistical principles.

Diffusion models can also be used to generate multi-view optical illusions, where a single set of features satisfies the different text prompts after performing specific transformations like rotations or flips[\[7\]](#).

A study from Ngo et al. (2023)[\[8\]](#) showed that Large Vision-Language models can also be reliably deceived by lightness and geometric illusions. Their research revealed that although it understands the illusion, the visual encoder still falls for the trap (perceptual failure).

3.5. The Research Gap: From 2D to 3D Representations

Despite all the current research, a significant gap still remains, The state-of-the-art main focus point is 2D visual images, now that it has been proven empirically that the AI models can replicate human-responses like brightness and color shifts by Gomez-Villa and Ngo, the next step is to understand how these biases manifest in a 3 dimensional space reconstruction.

4. Methodology and Work Plan

4.1. Methodology

This research adopts an Agile methodology, more specifically the Scrum framework. This methodology is based on a sprint methodology, where the workload is organized into bi-weekly Sprints. This ensures that each phase has a concrete manageable objective.

While Scrum traditionally relies on team synchronization, this TFG has adapted into daily self-check-ins to monitor progress and identifying technical bottlenecks early. After each sprint, a retrospective session is conducted with the project tutor to review the achieved goals and receive their feedback.

The choice of this methodology was driven by the need of flexibility, since researching the state-of-the-art AI models are highly unpredictable, this flexibility allows some pivoting or refinement in the experimental setup.

Finally, this framework allows for continuous refinement thanks to the feedback obtained throughout the different deliverables.

4.2. Work Plan (Timeline)

Sprint	Weeks	Dates (2026)	Technical Objectives & Tasks	Milestone
S1	1 - 2	Feb 9 - Feb 22	Project definition: Initial literature review. And dataset taxonomy.	Start of the Project
S2	3 - 4	Feb 23 - Mar 8	Technical setup: Environment configuration (Python/PyTorch) and loading Depth Anything V1-V3 weights.	-
S3	5 - 6	Mar 9 - Mar 22	Dataset curation: Final selection of illusory stimuli and writing the initial documentation.	Initial Report (Week 6)

S4	7 - 8	Mar 23 - Apr 5	Synthetic Stimuli Engineering: Development of scripts to generate controlled versions of the "Corridor Illusion". Implementation of variable parameters (swapping cylinders for cubes, spheres, etc.).	-
S5	9 - 10	Apr 6 - Apr 19	Systematic Stress Testing: Running Depth Anything V1, V2, and V3 through the synthetic library. Identifying "failure triggers" by comparing actual geometry vs. AI predicted depth.	-
S6	11 - 12	Apr 20 - May 3	Generative 3D Bias analysis: Evaluating 3D generative models (e.g., Hunyuan 2.1) to identify systemic geometric biases and reconstruction "hallucinations" when prompted with illusory stimuli..	-
S7	13 - 14	May 4 - May 17	Quantitative Metric Investigation: Researching and defining a standardized protocol of metrics to evaluate the AI outputs.	Progress Report (Week 13)
S8	15 - 16	May 18 - May 31	Biological Benchmarking: Analyzing AI "hallucinations" with biological mechanisms such as the hierarchical V1 cortical organization (Simple vs. Complex cells) described by Hubel & Wiesel among others.	-

S9	17 - 18	Jun 1 - Jun 14	Synthesis: Drafting the final thesis in article format (max. 12 pages) and discussing results with the tutor. Also, confirming or refuting research hypotheses.	Final Proposal (Week 18)
S10	19 - 20	Jun 15 - Jun 28	Closure: Final polishing of the TFG Dossier and preparation of the public defense slides.	Final Delivery (Jun 28)

4.3. Resource requirements

The execution of this project requires a high-performance computational environment to handle the inference of Large Vision Models. The resources are categorized into hardware and software components.

4.3.1. Hardware requirements

The primary technical constraint of this research is the VRAM and system memory required to run the different foundation models, such as the different versions of the Depth Anything. To ensure a precise comparison while maintaining adequate results, the base version[\[9\]](#) of the models was selected.

The most resource-intensive model used is Depth Anything V3, which uses a DinoV2 backbone[\[10\]](#), with 0.12B parameters.

The following hardware was utilized during this project:

1. CPU: Intel(R) Core(TM) i9-9900K @ 3.60 GHz.
2. GPU: NVIDIA GeForce RTX 3070 with 8 GB of dedicated VRAM.
3. RAM: 32 GB of System RAM.
4. Storage: 1 TB NVMe SSD (WDC WDS100T2B0C)

4.3.1.1 External & Cloud Computing Resources

Certain experimental phases involve high-parameter generative models, such as Hunyuan 2.1, which has 17 billion parameters[\[11\]](#), which significantly exceed the 8GB VRAM capacity of the local GPU. To address this hardware constraint, the project utilizes Hugging Face's ZeroGPU for Spaces[\[12\]](#).

Although the platform operates under a very restricted quota, it resets daily, allowing the temporary use of high-end hardware such as NVIDIA H200, for diverse AI tasks. In this case, the 3D mesh reconstruction pipeline[13].

4.3.2. Software and Frameworks

This development is built on a Windows distribution using a Python environment 3.10. With the following core libraries:

1. PyTorch (CUDA version): The primary framework used for tensor computation and GPU acceleration. Results in low-latency inference.
2. HuggingFace transformers: Used to access the optimized weights and configurations of the Depth Anything family.
3. OpenCV and Matplotlib: For the generation of visual comparison and image normalization.
4. Numpy: for the extraction of depth values and spatial analysis.

5. References

Foundational Theory & psychology

- [1] Gregory, R. L. (1997). Knowledge in perception and illusion. *Philosophical Transactions of the Royal Society of London. Series B: Biological Sciences*, 352(1358), 1121–1128.
<https://doi.org/10.1098/rstb.1997.0095>
- [2] Hubel, D. H., & Wiesel, T. N. (1962). Receptive fields, binocular interaction and functional architecture in the cat's visual cortex. *The Journal of Physiology*, 160(1), 106–154. DOI: <https://physoc.onlinelibrary.wiley.com/doi/10.1113/jphysiol.1962.sp006837>
- [3] Kitaoka, A., Ashida, H., & Murakami, I. (2005). Does the peripheral drift illusion generate illusory motion in depth? *Ritsumeikan University*. <https://www.psy.ritsumei.ac.jp/akitaoka/PDIstereoM.pdf>
- [4] Conway, B. R., Kitaoka, A., Yazdanbakhsh, A., Pack, C. C., & Livingstone, M. S. (2005). Neural basis for a powerful static motion illusion. *The Journal of Neuroscience*, 25(23), 5651-5656. <https://www.jneurosci.org/content/jneuro/25/23/5651.full.pdf>
- [5] Sugihara, K. (2015). Ambiguous Cylinders: a New Class of Impossible Objects. *Computer Aided Drafting, Design and Manufacturing (CADDM)*, Vol. 25, No. 4, pp. 19-25..

Ivan Martín Campoy

1673246

[6] Gomez-Villa, A., Twardowski, B., Wang, K., Malo, J., Parraga, C. A., van de Weijer, J., & Vazquez-Corral, J. (2025). The Art of Deception: Color Visual Illusions and Diffusion Models. Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR).

[7] Geng, D., Park, I., & Owens, A. (2024). Visual anagrams: Generating multi-view optical illusions with diffusion models.

[8] Ngo, J., Sankaranarayanan, S., & Isola, P. (2023). Is CLIP Fooled by Optical Illusions? ICLR Tiny Papers.

Models & Technical Resources

[9] Depth Anything Team (2025). Depth Anything V3 Base Model (DA3-BASE). Hugging Face Model Hub. Available at: <https://huggingface.co/depth-anything/DA3-BASE>

[10] Depth Anything Project (2025). Depth Anything V3: Official Documentation and Technical Showcase. Available at: <https://depth-anything-3.github.io>

[11] Tencent Hunyuan Team (2026). HunyuanImage-2.1: Next-Generation DiT-based Image Synthesis. Hugging Face Model Hub. Available at: <https://huggingface.co/tencent/HunyuanImage-2.1>

[12] Hugging Face Documentation (2026). ZeroGPU Spaces: Managing High-Performance Inference for Large-Scale Models. Available at: <https://huggingface.co/docs/hub/en/spaces-zerogpu>

[13] Tencent Hunyuan Team (2026). Hunyuan3D-2.1: High-Fidelity 3D Mesh Generation from Illusory Stimuli. Hugging Face Spaces. Available at: <https://huggingface.co/spaces/tencent/Hunyuan3D-2.1>

6. Annex

Impossible objects

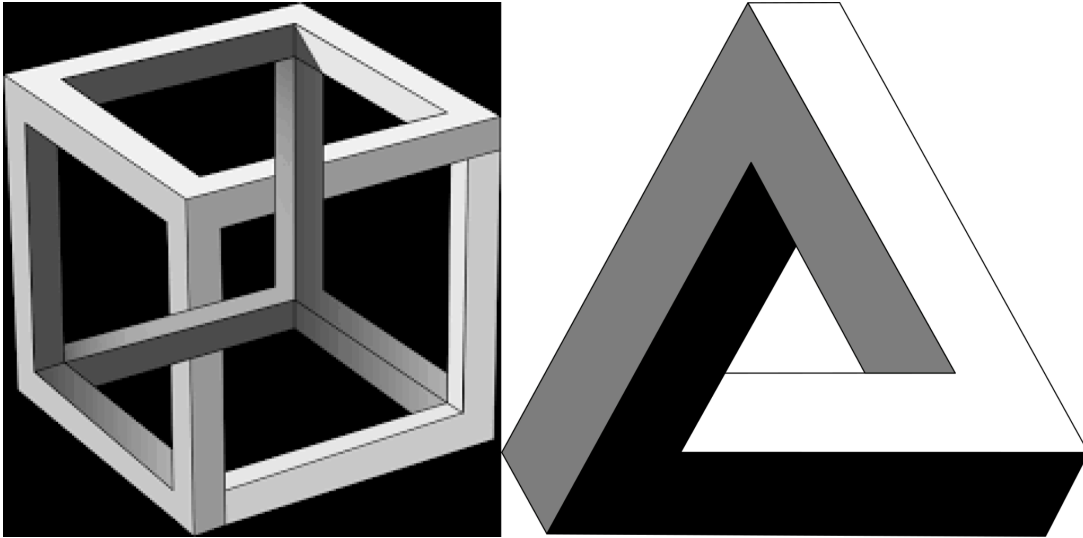


Figure 1. Escher's Cube (left) and Penrose Triangle (right)

Forced Perspective

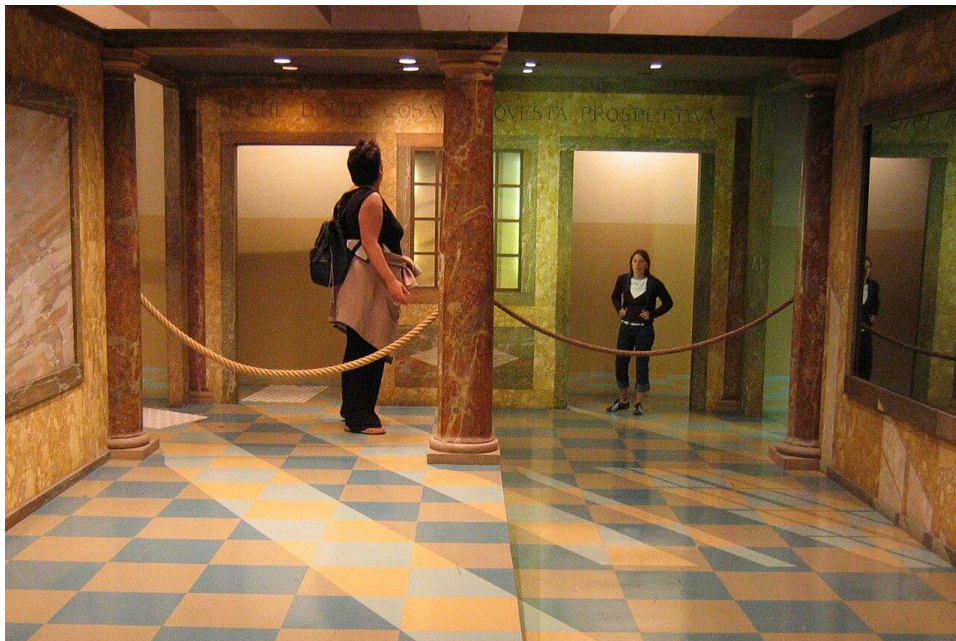


Figure 2. Ames Room

Ivan Martín Campoy

1673246

Depth Ambiguity

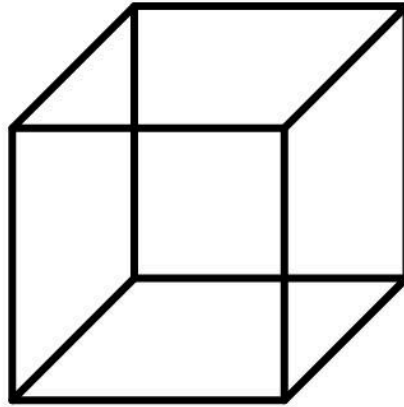


Figure 3. Necker's Cube

Converging Lines (Linear perspective)

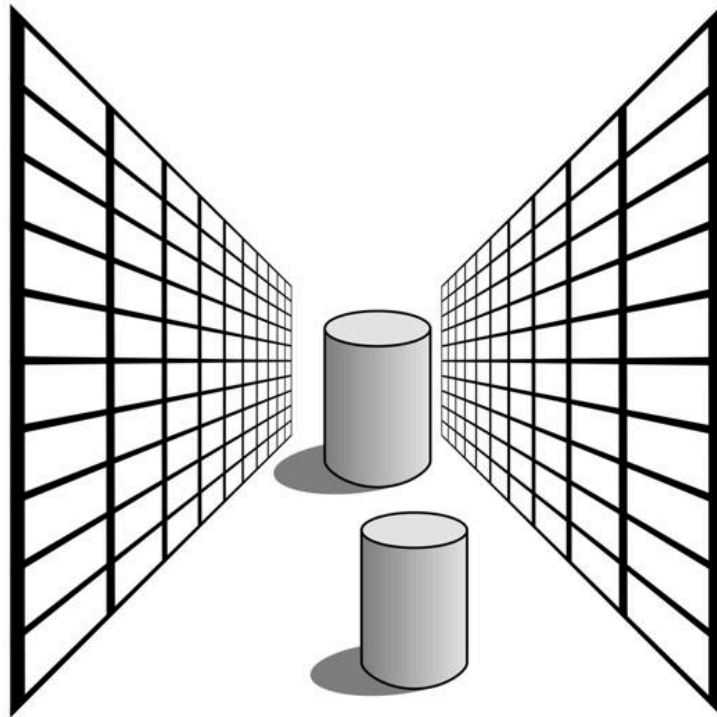


Figure 4. Corridor's Illusion

Ivan Martín Campoy

1673246

Shading & Illumination (Shape-from-shading)



Figure 5. Gathering for Gardner's Dragon Illusion